20285 Advanced Macroeconomics

Problem set 4

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Exercise 1: Optimal price setting in the Taylor model

In this exercise, we explore the Taylor model of staggered price setting, where each firm resets its price for N periods. In each period, a fraction $\frac{1}{N}$ of firms reset their prices, while the remaining firms keep their prices fixed. Each firm i produces a differentiated good with technology:

$$Y_t(i) = A_t N_t(i),$$

where A_t represents the level of technology, evolving exogenously as $a_t = \log A_t$ with a stationary stochastic process. A firm resetting its price P_t^* in period t seeks to maximize its expected profits over N periods:

$$\max_{P_t^*} \sum_{k=0}^{N-1} \mathbb{E}_t \left\{ Q_{t,t+k} \left(\frac{1}{P_{t+k}} \right) \left(P_t^* Y_{t+k|t} - \Psi_{t+k}(Y_{t+k|t}) \right) \right\}$$

subject to:

$$Y_{t+k|t} = \left(\frac{P_t^*}{P_{t+k}} \right)^{-\epsilon} C_{t+k},$$

where $Q_{t,t+k} = \beta^k \left(\frac{C_{t+k}}{C_t} \right)^{-\sigma}$ is the stochastic discount factor, $\Psi_{t+k}(Y_{t+k|t})$ are the nominal cost, and C_{t+k} is aggregate consumption in period $t+k$.

1. Show that the optimal reset price P_t^* must satisfy the first-order condition:

$$\sum_{k=0}^{N-1} \mathbb{E}_t \left\{ \left(\frac{1}{P_{t+k}} \right) Q_{t,t+k} Y_{t+k|t} \left(P_t^* - \mathcal{M}\psi_{t+k} \right) \right\} = 0,$$

where $\psi_{t+k} = \Psi'(\cdot)$ is the nominal marginal cost and $\mathcal{M} = \frac{\epsilon}{\epsilon-1}$.

[Hint: Set up the first-order condition by differentiating the profit function with respect to P_t^*. Remember to substitute the demand function and use the stochastic discount factor for nominal payoffs.]

Solution.

2. Derive the log-linearized version of the optimal price-setting rule around a zero-inflation steady state:

$$p_t^* = \mu + \sum_{k=0}^{N-1} \omega_k \mathbb{E}_t \left\{ \varphi_{t+k} \right\},$$

where $\omega_k = \frac{\beta^k(1-\beta)}{1-\beta^N}$, $\varphi_{t+k} = \log \psi_{t+k}$, and $\mu = \log \mathcal{M}$.

Interpret this condition.

[Hint: Approximate the optimal price around a zero-inflation steady state by log-linearizing the price-setting rule. This will involve taking logarithms and expectations.]

Solution.

3. How does the Taylor model's price-setting rule differ from the one generated by the Calvo model? Discuss the economic intuition behind these differences, focusing on the implications for inflation persistence. How do they compare in terms of tractability?

Solution.

4. Recalling that the aggregate price index P_t is defined by:

$$P_t = \left(\int_0^1 P_t(i)^{1-\epsilon} \, di \right)^{\frac{1}{1-\epsilon}},$$

show that, around a zero-inflation steady state, the (log) price level will satisfy:

$$p_t = \frac{1}{N} \sum_{k=0}^{N-1} p_{t-k}^*.$$

[Hint: Argue why we have symmetric pricing for firms within each cohort and derive the relationship between the log price level p_t and the optimal reset prices p_{t-k}^*.]

Solution.

Exercise 2: The Phillips Curve with Price Indexation

In the New Keynesian model, there is a probability θ that a firm will not be able to change its nominal price in a given time period. Firms that do have an opportunity to change price at time t choose a reset price p_t^* according to:

$$p_t^* = (1 - \beta\theta) \sum_{\ell=0}^{\infty} (\beta\theta)^\ell \mathbb{E}_t [p_{t+\ell} + \gamma \widehat{mc}_{t+\ell}],$$

where β is the discount factor, p_t is the price level, $\widehat{mc}_t$ is average marginal cost, and $\gamma = \frac{1-\alpha}{1-\alpha+\alpha\epsilon}$ is a positive coefficient. All variables are given as log deviations from their steady-state values. The general price level p_t is determined by:

$$p_t = (1 - \theta) \sum_{\ell=0}^{\infty} \theta^\ell p_{t-\ell}^*.$$

1. Provide an intuitive explanation of the two equations above.

   Solution.

2. Show that the reset-price and price-level equations are equivalent to:

$$p_t^* = \beta\theta\mathbb{E}_t p_{t+1}^* + (1-\beta\theta)(p_t + \gamma\widehat{mc}_t),$$

   and

$$p_t = \theta p_{t-1} + (1-\theta)p_t^*.$$

   Solution.

3. By combining the two equations, derive the New Keynesian Phillips curve for inflation $\pi_t = p_t - p_{t-1}$:

$$\pi_t = \beta\mathbb{E}_t\pi_{t+1} + \kappa\widetilde{y}_t,$$

   using that $\widehat{mc}_t = \left(\sigma + \frac{\alpha+\phi}{1-\alpha}\right)\widetilde{y}_t$, and where $\kappa = \gamma\left(\sigma + \frac{\alpha+\phi}{1-\alpha}\right)\left(\frac{(1-\theta)(1-\beta\theta)}{\theta}\right)$

   Solution.

Next suppose that when a firm does not have an opportunity to choose a new reset price optimally at time t (probability θ), the firm adjusts its old nominal price in line with any past inflation π_{t-1} that has occurred.[1] This means the equation for the price level is now:

$$p_t = \theta(p_{t-1} + \pi_{t-1}) + (1-\theta)p_t^*.$$

When choosing the reset price p_t^*, firms take account of the indexation of prices to past inflation. Hence the new equation for the reset price is:

$$p_t^* = (1-\beta\theta)\sum_{\ell=0}^{\infty}(\beta\theta)^\ell\mathbb{E}_t[p_{t+\ell} - (p_{t+\ell-1} - p_{t-1}) + \gamma\widehat{mc}_{t+\ell}].$$

4. Explain the intuition for the new reset-price equation.

   Solution.

5. Derive the new Phillips curve. Comment on any differences relative to the standard New Keynesian Phillips curve.

   [Hint: Start by putting the reset-price equation in a simpler recursive form.]

   Solution.

[1] For a more detailed treatment, see for example "Trend Inflation, Indexation, and Inflation Persistence in the New Keynesian Phillips Curve" by Cogley and Sbordone (AER 2008).

Exercise 3: The Phillips Curve

Part 3.1: The Classic Phillips Curve

For this exercise, download the following series at quarterly frequency from 1960:Q1 through 2025:Q4 from FRED:

- u_t = *Unemployment Rate* (Unit: Percent, Seasonally Adjusted);
- π_t = *Personal Consumption Expenditures: Chain-type Price Index* (Unit: Percent Change from Year Ago, Seasonally Adjusted);

1. Construct a series for expected price inflation, measured as $\widetilde{\mathbb{E}}_t[\pi_{t+1}] = \frac{1}{4}(\pi_{t-1} + \pi_{t-2} + \pi_{t-3} + \pi_{t-4})$. Briefly interpret and comment on this measure of inflation expectations.

   Solution.

   The backward-looking measure $\tilde{E}_t[\pi_{t+1}] = \frac{1}{4}(\pi_{t-1}+\pi_{t-2}+\pi_{t-3}+\pi_{t-4})$ embodies an *adaptive expectations* hypothesis: agents form their forecast of next-period inflation by simply averaging the four most recent quarterly inflation outcomes. Because it is a four-quarter moving average, this series is mechanically smoother than actual inflation—it dampens transitory spikes and responds only gradually to persistent shifts. The measure implicitly assumes that agents use *only* past own-inflation to forecast, ignoring all other potentially informative variables (monetary policy announcements, survey data, commodity prices, etc.). This is a strong assumption, and it means the measure will systematically lag behind turning points in actual inflation and may fail to capture the forward-looking component of expectations that drives pricing decisions.

2. Create a scatterplot of u_t (x-axis) and $(\pi_t - \widetilde{\mathbb{E}}_t[\pi_{t+1}])$ (y-axis) for the entire period. Fit a simple OLS regression line and plot it in the same graph. Briefly describe your findings. What is $(\pi_t - \widetilde{\mathbb{E}}_t[\pi_{t+1}])$ measuring?

   Solution.

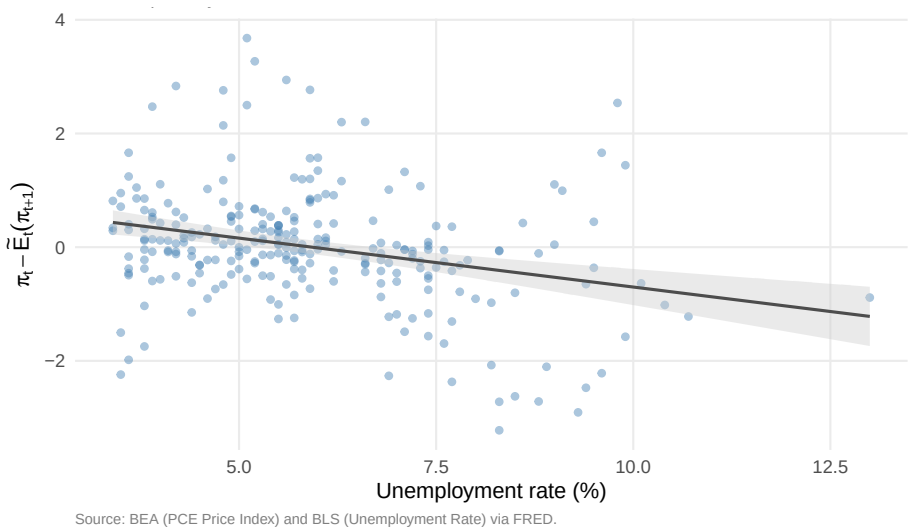


Figure 3-1. The accelerationist Philips curve (full sample); N = 260 quarterly observations.

The dependent variable $y_t = \pi_t - \tilde{E}_t[\pi_{t+1}]$ measures the *inflation surprise*—the deviation of realised inflation from backward-looking expected inflation. A negative slope coefficient on u_t is consistent with the Phillips curve: when unemployment is high, inflation tends

to come in *below* expectations. The full-sample OLS fit ($\hat{\kappa} = -0.172$, HC1 SE $= 0.044$; $R^2 = 0.081$, $N = 260$) confirms a small, statistically significant negative slope. The economic magnitude is modest and the R^2 low, reflecting the well-known instability of the reduced-form Phillips curve: much of the variation in inflation surprises is driven by supply shocks, expectation shifts, and regime changes that a simple bivariate regression on unemployment cannot capture.

3. Repeat the exercise from above for the following subperiod:

   a. 1960:Q1 – 1986:Q4 (blue)
   b. 1987:Q1 – 2007:Q4 (green)
   c. 2008:Q1 – 2019:Q4 (red)
   d. 2020:Q1 – 2025:Q4 (black)

   Compare the scatterplots and slopes over time and discuss.

   Solution.

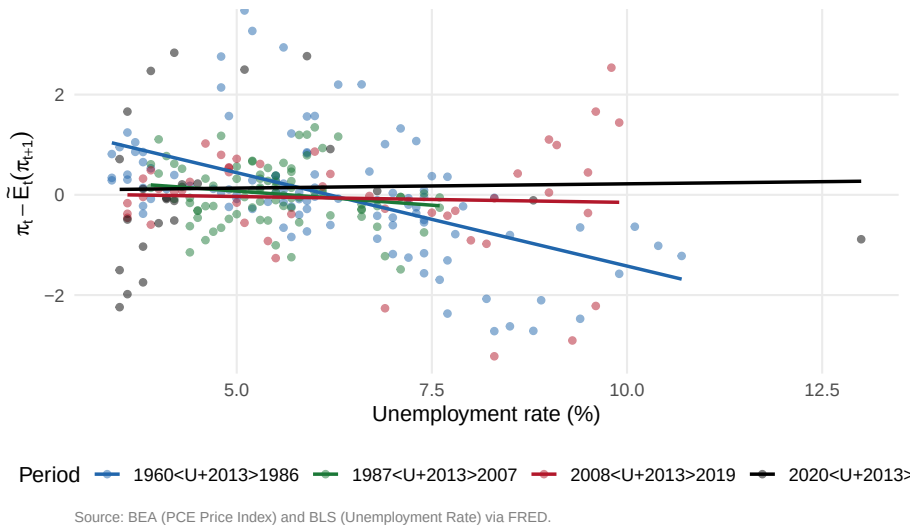


Figure 3-2. The Philips curve across subperiods.

Period	$\hat{\kappa}$	HC1 SE	R^2	N
1960–1986	−0.373	0.050	0.294	104
1987–2007	−0.112	0.067	0.031	84
2008–2019	−0.024	0.090	0.002	48
2020–2025	+0.017	0.116	0.001	24

Table 3-1. Phillips-curve slope by subperiod. OLS of $\pi_t - \tilde{E}_t[\pi_{t+1}]$ on u_t, with HC1-robust standard errors.

The subperiod analysis (Table 3-1) reveals a striking *flattening* of the Phillips curve over time. During 1960–1986—a period encompassing both the Great Inflation and the Volcker disinflation—the slope is steeply negative ($\hat{\kappa} \approx -0.37$, $R^2 \approx 0.29$), reflecting large swings in both unemployment and inflation surprises. The Great Moderation era (1987–2007) shows a much flatter relationship ($\hat{\kappa} \approx -0.11$): well-anchored expectations meant that unemployment fluctuations had diminished pass-through to inflation. The post-GFC period (2008–2019) is flatter still ($\hat{\kappa} \approx -0.02$, R^2 essentially zero), consistent with the "missing

disinflation" puzzle—unemployment rose dramatically during and after the Great Recession, yet inflation barely fell below expectations, suggesting that expectations anchoring (possibly to the Fed's 2% target) rendered the traditional trade-off nearly moot. The COVID era (2020–2025) is harder to read: although the *scatter* spans an extraordinary range—from the 2020:Q2 unemployment spike paired with deflationary surprises to the 2021–2022 inflation surge at very low u_t—the OLS slope is statistically indistinguishable from zero ($\hat{\kappa} \approx 0.02$, HC1 SE ≈ 0.12, $N = 24$), because supply-driven outliers dominate a short and noisy sample.

This pattern—flat OLS over the stable post-2008 period, with the COVID episode dominated by outliers that the linear-in-u specification cannot accommodate—foreshadows the non-linear specification we examine in Question 9, where the Benigno and Eggertsson (2024) "slanted-L" framework formalises the conditions under which the relationship switches between regimes.

Part 3.2: Household expectations

4. Read and summarize the key question and findings of "Is the Phillips Curve Alive and Well after All? Inflation Expectations and the Missing Disinflation" by Coibion and Gorodnichenko (AEJ: Macro 2015).

Solution.

The central question addressed by Coibion & Gorodnichenko (2015) is the "missing disinflation" puzzle: why did US inflation not fall much further (or enter deflation) during the Great Recession of 2008–2011, despite the massive spike in unemployment and economic slack? Traditional Phillips curve models, which relied on backward-looking expectations or forecasts by professionals, predicted a much sharper disinflation.

Their key finding is that the Phillips curve is actually "alive and well" once you use the *correct* measure of inflation expectations. They document that household inflation expectations (as measured by the Michigan Survey of Consumers) closely track salient prices, particularly oil and gasoline. During the 2009–2011 period, a surge in global oil prices pushed household inflation expectations higher, even as the broader economy remained in a severe slump. When these survey-based expectations are used in an expectations-augmented Phillips curve, the implied negative inflation surprise ($\pi_t - E_t^{MSC}$) matches the high unemployment rate perfectly. This fully explains the missing disinflation without needing to assume that the underlying slope of the Phillips curve had flattened.

A complementary piece of evidence underpins the entire interpretation. Coibion and Gorodnichenko (2015) exploit the New Zealand business survey—one of the few datasets that directly measures *firm* inflation expectations—and document that firms' expectations co-move strongly with household survey forecasts and only weakly with professional forecasts. This finding is what licenses the use of MSC data as a proxy for the expectations relevant to firm price-setting, and is therefore central to the paper's claim that the Phillips curve is "alive and well" once the right expectations measure is used.

In this exercise, we focus on the third potential explanation for the missing disinflation discussed in Section IV of the paper. Begin by downloading the following time series:

⇒ $\mathbb{E}_t^{MSC}[\pi_{t+4}]$ = *Household inflation forecast, Median 1-year ahead* (Source: University of Michigan - Survey of Consumers, Table 32, time span: 1978:Q1-2025:Q4).

Next, we will estimate the Phillips curve using household forecasts. Specifically, we consider a Friedman (1968)-type expectations-augmented Phillips curve:

$$\pi_t - \mathbb{E}_t[\pi_{t+1}] = c + \kappa u_t + v_t, \tag{3.1}$$

as in equation (1) of Coibion and Gorodnichenko (2015). We are interested in estimating κ, the slope of the Phillips curve. To do so, we will implement different measures of inflation expectations, $\mathbb{E}_t[\pi_{t+1}]$.

5. First, plot, describe and compare the time series of the two measures of inflation expectations, $\widetilde{\mathbb{E}}_t[\pi_{t+1}]$ and $\mathbb{E}_t^{MSC}[\pi_{t+4}]$). Why is it useful to use the median (instead of the mean for example) for household inflation expectations?

Solution.

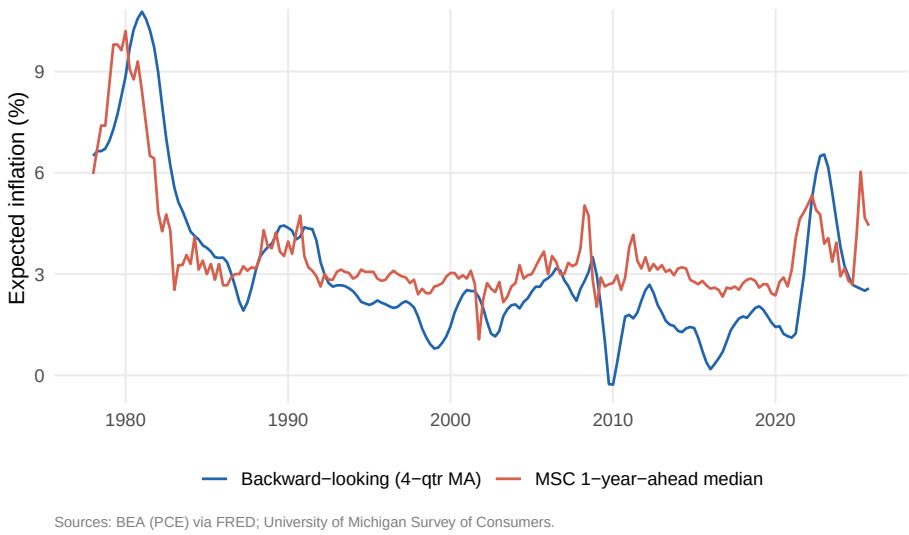


Figure 3-3. Two measures of inflation expectations: the backward-looking 4-quarter moving average $\tilde{E}_t[\pi_{t+1}]$ and the Michigan Survey of Consumers median 1-year-ahead expectation $E_t^{MSC}[\pi_{t+4}]$, 1978:Q1–2025:Q4.

Figure 3-3 overlays the two series on a common axis from 1978:Q1 onward. Several key differences emerge between the backward-looking measure $\tilde{E}_t[\pi_{t+1}]$ and the Michigan Survey of Consumers (MSC) 1-year-ahead median expectation $E_t^{MSC}[\pi_{t+4}]$:

- **Level:** The MSC median expectation tends to run *above* the backward-looking measure, particularly during the low-inflation period after 2000. Households systematically perceive and expect higher inflation than is recorded by the PCE index, a well-documented finding in the survey literature.
- **Volatility:** The backward-looking measure, being a four-quarter moving average of actual inflation, is mechanically smooth. The MSC median is somewhat more volatile and can move abruptly in response to salient price signals (e.g., gasoline prices).
- **Co-movement with actual inflation.** Both series co-move with realised inflation at low frequencies, but the MSC measure diverges notably during the post-GFC period. Coibion and Gorodnichenko (2015) document that between 2009 and 2013, MSC household expectations rose from roughly 2.5% to a peak of about 5% in 2011 and remained in the 3–4% range thereafter, while financial-market and SPF measures stayed anchored near 2%. This wedge—driven largely by gasoline prices—is the empirical fact Coibion and Gorodnichenko (2015) exploit to resolve the missing-disinflation puzzle.

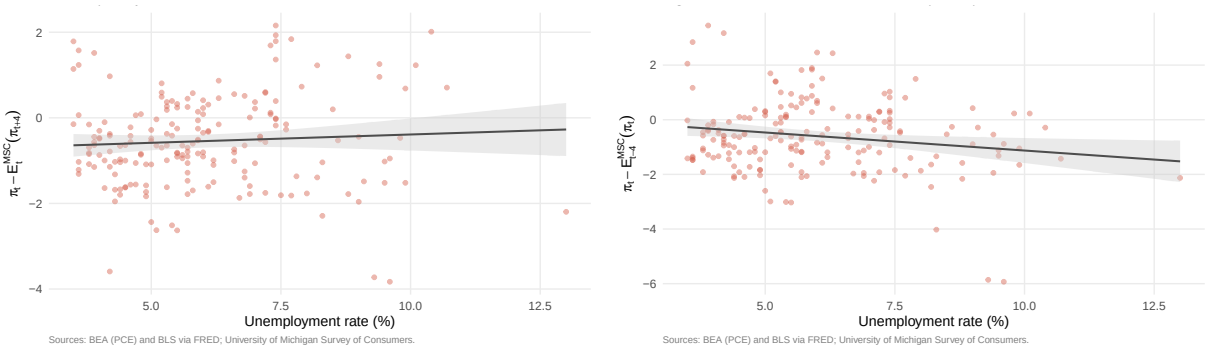
The *median* (rather than the mean) is preferred for household expectations because survey response distributions are typically right-skewed and fat-tailed. A small fraction of respondents consistently report extreme values (e.g., 20–50% expected inflation), which

disproportionately inflate the mean. The median is robust to these outliers and provides a more representative summary of the central tendency of household beliefs.

6. Next, estimate (3.1) assuming that $\mathbb{E}_t[\pi_{t+1}] = \mathbb{E}_t^{MSC}[\pi_{t+4}]$. Create a scatter plot and include your estimated regression line, as in Figure 6 Panel B of the paper.

   Relate your findings to the results of Coibion and Gorodnichenko (2015) and discuss.

   Solution.



(a) Contemporaneous: $\pi_t - E_t^{MSC}[\pi_{t+4}]$. **(b)** Aligned: $\pi_t - E_{t-4}^{MSC}[\pi_t]$.

Figure 3-4. Phillips curve with household expectations. Panel (a) uses the contemporaneous MSC forecast; Panel (b) uses the forecast made one year earlier (lag-4), aligning the forecast horizon with the realisation it is meant to predict.

Replacing the backward-looking measure with the Michigan Survey forecast yields a markedly different scatterplot from Question 2. The cloud of observations is shifted downward—most surprises are negative—but its tilt is essentially flat: with the contemporaneous specification the fitted slope is small, of indeterminate sign, and statistically indistinguishable from zero ($\hat{\kappa} = +0.039$, HC1 SE $= 0.058$; $R^2 = 0.004$, $N = 192$), with a 95% confidence band that crosses the horizontal axis across nearly the entire range of u_t.

Two features of the scatter deserve attention before any structural interpretation. The first is a *level shift*: $\pi_t - E_t^{MSC}[\pi_{t+4}]$ averages around -0.6 over the sample, reflecting the well-documented tendency for household survey expectations to run persistently above realised PCE inflation. The second is the absence of a clear conditional pattern: even at high unemployment, where the standard Phillips curve predicts large negative surprises, the cloud is diffuse rather than concentrated in the lower-right region of the plot.

This is precisely the picture Coibion and Gorodnichenko (2015) emphasise. MSC respondents weight salient consumer prices—gasoline and food—far more heavily than the weights these items receive in the broad PCE deflator. Whenever commodity prices move independently of the macroeconomic cycle (as in 2009–2011, when oil prices recovered while broader prices remained slack), MSC expectations decouple from realised inflation in a way that mechanically generates negative surprises unrelated to the unemployment rate. The apparent flattening of the curve is therefore not evidence that the structural slack–inflation relationship has weakened; it is a measurement artifact induced by the expectations proxy.

The 2020–2022 observations work against the post-GFC pattern in the opposite direction. With unemployment falling rapidly and realised inflation overshooting the MSC forecast, this period contributes positive surprises at low u_t, pulling the fitted line further toward a flat or mildly positive slope. The plot thus combines two distinct identification problems: a measurement bias in the expectations regressor, and a small number of high-leverage post-pandemic observations.

A final remark concerns timing. The dependent variable as specified in Panel (a), $\pi_t - E_t^{MSC}[\pi_{t+4}]$, is not a properly aligned conditional surprise: π_t is current year-on-year inflation while $E_t^{MSC}[\pi_{t+4}]$ is a forecast for a horizon four quarters out, so the realisation and the forecast horizon do not match. Panel (b) substitutes $E_{t-4}^{MSC}[\pi_t]$—the median forecast made one year earlier of the YoY inflation rate that has just realised—which aligns the forecast horizon with the outcome it predicts. The slope flips from a small positive value to a visibly negative one ($\hat{\kappa} = -0.132$, HC1 SE $= 0.059$; $R^2 = 0.032$, $N = 188$). This sensitivity is itself a useful diagnostic: a non-trivial part of the apparent "flattening" in Panel (a) is mechanical mis-alignment between the realisation and the forecast, not evidence on the structural Phillips slope. Identifying that slope requires more than swapping out the expectations measure—an issue we develop in Question 7.

7. Briefly discuss the OLS estimation of the slope of the Phillips curves in 3.2, 3.3. and 3.6.

   [Hint: Specifically, consider the identification problem of simultaneity bias in the given case. You may want to have a look at Section 2 and, more importantly, Section 3 of McLeay and Tenreyro (2020, NBER Macroeconomics Annual *34).]*

   Solution.

   A fundamental identification problem afflicts all the OLS regressions estimated above (in Questions 2, 3, and 6): **simultaneity (reverse causality)**. Inflation expectations and unemployment are *jointly determined*—aggregate demand shocks simultaneously raise inflation (or expectations thereof) and lower unemployment, while monetary policy responses to inflation and employment conditions shift the economy along both dimensions. OLS of inflation on unemployment therefore conflates movements *along* the Phillips curve (driven by supply shocks, which we want to identify) with shifts *of* the Phillips curve (driven by demand shocks and policy responses).

   McLeay and Tenreyro (2020), in *NBER Macroeconomics Annual* 34, formalise this argument. They show that when the central bank conducts *optimal monetary policy*—adjusting interest rates to stabilise inflation in response to unemployment gaps—the policy reaction function induces a spurious positive correlation between inflation and unemployment in equilibrium. This endogeneity biases the OLS estimate of the Phillips curve slope toward zero (or even toward a positive sign). The PC is identified *only* off supply-side shocks or when an instrumental variable for monetary policy is available.

   Relating this back to our estimates:

   - **Question 2 (Full sample, backward-looking):** The near-zero slope is *most* susceptible to this bias. Backward-looking expectations move slowly and essentially proxy for lagged inflation, so the regression captures a mix of demand- and supply-driven variation, with the Fed's stabilisation policy washing out much of the PC signal.
   - **Question 3 (Subperiods):** The subperiod estimates are similarly biased. The observed "flattening" during the Great Moderation (1987–2007) may partly reflect a more aggressive inflation-targeting regime by the Fed, which offset demand shocks more effectively, rather than a true structural flattening of the underlying Phillips curve.
   - **Question 6 (MSC expectations):** While somewhat less biased because household survey expectations capture *forward-looking* information that partially controls for demand-shock-induced expectation shifts, simultaneity remains. Demand booms raise both MSC expectations and employment, confounding the true slope estimate.

The only clean identification requires either an *instrument* for monetary policy (e.g., narrative shocks à la Romer & Romer) or an instrument for slack (e.g., Bartik-style local labour-market instruments, as in Gitti 2024).

Part 3.3: Non-linear Phillips curves

8. Read "*The Slanted-L Phillips Curve*" by Benigno and Eggertsson (2024) (AEA Papers and Proceedings, 2024). Briefly summarize the authors' key arguments.

   Solution.

   Benigno & Eggertsson (2024) argue that the standard *linear* New Keynesian Phillips curve misses a critical feature of the inflation–labour-market-tightness relationship: it is **non-linear, exhibiting a convex (or kinked) shape**. Specifically, they document a "slanted-L" pattern in the data:

   - **At low tightness** (high unemployment, low V/U), additional slack has *little* disinflationary effect—the Phillips curve is nearly flat. This explains the "missing disinflation" of 2009–2015: despite very high unemployment, inflation did not fall substantially because the economy was on the flat portion of the curve.
   - **At high tightness** (low unemployment, high V/U), small increases in tightness generate *large* inflation increases—the curve becomes steep, nearly vertical. This explains the rapid inflation surge of 2021–2022: the economy quickly moved onto the steep portion of the curve.

   The key theoretical mechanism is a *capacity constraint on production*: firms can substitute between vacancies and labour to adjust output, but as the labour market approaches full utilisation the marginal cost curve becomes very steep—effectively kinking—because additional output requires either bidding scarce workers away from other firms or accepting longer vacancy durations. Once this constraint binds, small increases in tightness map into large increases in marginal cost and hence in price-setting markups, generating the near-vertical arm of the slanted-L. Below the kink, marginal cost is roughly flat in tightness, so additional slack has little disinflationary bite. Search-and-matching frictions are a complementary force: when V/U is high, hiring takes longer and wage offers must rise to attract workers, reinforcing the cost-side amplification. Crucially, Benigno & Eggertsson argue that *no structural break or shift in the Phillips curve is needed* to explain both the "missing disinflation" and the "inflation surge"—both phenomena arise naturally from the non-linear shape of a *stable* underlying relationship.

9. Replicate Figure (1) of the paper for the US. To do so, simply perform the following steps:

   (a) Download quarterly data from the following series from Q1 2000 through Q4 2025 from FRED:
      - V_t = *Job Openings: Total Non-Farm* (Unit: Level in Thousands, Seasonally Adjusted);
      - U_t = *Unemployment Level* (Unit: Thousands of Persons, Seasonally Adjusted);

   (b) For each period, compute the V_t/U_t ratio. Finally, create a scatter plot of V_t/U_t (x-axis) and π_t (y-axis).

   (c) Briefly describe your findings and discuss.[2]

[2] For a more detailed descriptive background see Benigno and Eggertson(2024) and Benigno and Eggertson(2024).

Solution.

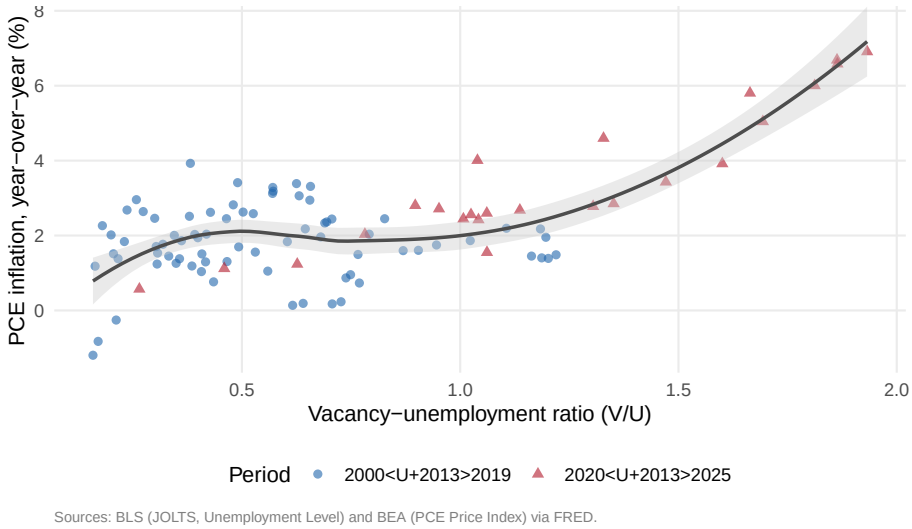


Figure 3-5. The non-linear Phillips curve.

The scatterplot reveals the characteristic **"slanted-L" shape** described by Benigno and Eggertsson (2024). For V/U ratios below roughly 1, the relationship is essentially flat: across the 2000–2019 expansion, the GFC trough, and the late-cycle 2018–2019 quarters where V/U first crossed above one with inflation still anchored around 2%, year-over-year PCE inflation hovers in a narrow 1–2.5% band regardless of tightness. The steep, near-vertical arm of the curve is essentially the **2021:Q2–2023:Q1 cluster**: every quarter in our sample with PCE inflation above 4% sits in this window, with V/U ratios between 1.04 and 1.93. Within this cluster a roughly 0.9-point increase in V/U (from 1.04 to 1.93 over 2021:Q2–2022:Q2) is associated with PCE inflation rising from 4.0% to 6.9%. The kink itself sits near $V/U \approx 1$—close to where Benigno and Eggertsson (2024) place it in their Figure 1.

This pattern contrasts sharply with the **linear Phillips curve**, which would predict a uniform, moderate negative relationship between slack and inflation across the entire range. The linear specification would substantially overpredict disinflation during the slack post-GFC period and underpredict inflation during the 2021–2022 boom. The non-linear relationship shows that inflation is relatively insensitive to tightness when the labour market has ample slack—firms can fill vacancies without large wage increases—but becomes highly sensitive once tightness pushes the economy past the kink and onto the capacity-constrained arm. As the post-2022 observations show, the unwinding of tightness brought inflation back down, tracing the curve in reverse: the 2024–2025 quarters are back at $V/U \approx 1$ with PCE inflation around 2.5%.

10. Lastly, read Giulia Gitti's Job Market Paper (JMP, 2024). Explain her approach of addressing potential endogeneity when relating labor market tightness and inflation. How does this relate to your findings in 3.8 and 3.9?

Solution.

Gitti (2024) addresses a key endogeneity concern that plagues the OLS scatterplot in Question 9: the vacancy–unemployment ratio V/U is endogenous to inflation because demand shocks raise both labour-market tightness and prices simultaneously. Her identification strategy uses a Bartik-style (shift-share) instrumental variable in which the "shifts" are national-level vacancy shocks in *tradable intermediate-input* industries—chosen because demand for these industries' output is determined by national rather than local

conditions—and the "shares" are pre-determined local industry employment weights. Interacting the two yields exogenous variation in local labour-market tightness that is plausibly orthogonal to local demand shocks. To further isolate demand-driven variation, she controls for the relative price of intermediate inputs and for productivity shocks in the final-goods sector.

This approach relates directly to Question 9 (and Question 7): the OLS scatterplot is subject to the **same simultaneity bias** discussed previously. During demand booms, both tightness and inflation rise *simultaneously*, inflating the apparent slope of the tightness–inflation relationship. Conversely, during recessions, both decline together. The OLS estimates therefore confound the causal effect of tightness on inflation with the common response of both variables to demand conditions.

Gitti's IV strategy recovers a **cleaner causal estimate** of the tightness–inflation relationship, which she finds to be:

(a) **Non-linear**, consistent with the slanted-L shape documented by Benigno & Eggertsson (2024);

(b) **Quantitatively different** from OLS—the IV estimates suggest the true causal relationship is steeper in the tight-market region than OLS implies (because OLS averages over demand- and supply-driven variation);

(c) **Robust** to controlling for local demand conditions and to various specification checks on the instrument's validity.

Her results thus provide rigorous causal support for the non-linear Phillips curve hypothesis and demonstrate that naive OLS estimates—whether using unemployment (Q2, Q5) or tightness (Q8)—are biased by simultaneity.

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